Anaïs Azouaoui

MSc student in Computer Science, seeking a 6-month research internship from March 2027

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Research interests: robot learning, imitation and reinforcement learning, vision-language models, representation learning

Education

Master's in Computer Science, Sorbonne University, Paris — *2025 – 2027 (expected)*
Algorithms, AI, Interactions & Decision (AI2D)

BSc in Computer Science, Sorbonne University, Paris — *2022 – 2025*

Research Experience

Interpretable ML for Intensive Care | *Research Internship* — Jun – Aug 2026
ISAE-Supaero - CHU de Toulouse, Toulouse — Supervisors: G. Nadizar, S. Cussat-Blanc, D. Wilson

- Modeled the severity of 4 respiratory disorders, as rated by clinicians, in 164 mechanically ventilated ICU patients.
- Built a **Cartesian Genetic Programming** engine with **NSGA-II** to evolve short, readable formulas.
- Obtained 2–3-variable formulas with test R^2 of 0.62–0.87, on par with the best black-box model (Extra Trees, 0.65–0.83).

Real-to-Sim Scene Generation | *Research Project* — Feb – May 2026
ISIR, Sorbonne University, Paris — Supervisors: S. Doncieux, E. Garrabe

- Designed a multi-step **LLM/VLM** pipeline converting a text prompt or a photo into a physically valid 3D **Genesis** scene.
- Back-projected **GroundingDINO** boxes with metric **Depth Anything V2**, cutting real-photo 3D error from 1.25 to 0.48 m.
- Fine-tuned **Phi-3-mini** with 4-bit **QLoRA** (Unsloth, TRL), raising spatial-relation F1 from 56% (GPT-4o) to 72%.
- Mapped free-text object names (synonyms, French or English) to 3D assets by **embedding retrieval** with **multilingual-E5**.
- Built 3,531 synthetic text–scene pairs filtered by an **LLM verifier**, and a **VLM loop** correcting the rendered scene.

Projects

Damage Recovery for Legged Robots — 2026 | Project page

- Reproduced **Intelligent Trial & Error** (ITE): learn hundreds of gaits in simulation, test the most promising after damage.
- Built gait repertoires for 3 simulated **Brax** robots with **MAP-Elites**, **PGA-ME** and **DCRL-ME** in **QDax** (**JAX**).
- Developed 2 new methods inspired by meta-learning and RMA: learned **Gaussian process** prior, sensor reading.
- Raised performance recovered in 5 trials from 31% to 46% on 3 robots; 90× fewer episodes than **TD3** fine-tuning.

RoboSimon: Memory in Visuomotor Imitation Policies — 2026 | Project page

- Built a Simon game in **MuJoCo** where the robot sees a sequence of 2 to 4 lit buttons and must repeat it from memory.
- Trained **ACT** and **Diffusion Policy** with a shared **transformer** memory, so that only the action head differs.
- Found ACT reaches 100% while Diffusion Policy needs 3× more training to go from 34–78% to 99.7%.
- Showed with **counterfactual tests** and **linear probes** that its memory failures came from under-training.

Vision-Language-Action Model Robustness — 2025 | Project page

- Evaluated **SmolVLA** on 3,688 **LIBERO** episodes under visual and arm perturbations: a 3° joint offset halves success.
- Paraphrased instructions (**metamorphic testing**) cut success from 80% to 23%; fixed by **sentence-encoder** retrieval.
- Searched with **MAP-Elites** for mixes of small perturbations, harmless alone, that fail 71% of held-out episodes.

Camera-Guided Navigation of a Self-Built Wheeled Robot — 2026 | Project page

- Built a **C++17** navigation stack with **EKF** state estimation, A* motion planning and **MPPI** model predictive control.
- Ran the same code in simulation and on hardware, fixing the **sim-to-real** gap (static friction, brownouts, camera latency).
- Reached 10/10 real goals at 2 cm; MPPI solved 143 of 145 simulated goals without collision, against 116 for Pure Pursuit.

Skills

Domains	Deep Learning, Reinforcement Learning, Computer Vision, LLMs/VLMs, NLP, Robot Learning, Motion Planning
Methods	Transformers, QLoRA, Embedding Retrieval, Imitation Learning, Evolutionary Computation, EKF, MPPI
Tools	Python, C++17, PyTorch, JAX, Hugging Face, OpenCV, scikit-learn, Simulation (Genesis, LeRobot), Git, Docker
Languages	French (native), English (proficient), Arabic (proficient), Japanese (basic)

Publications

[1] **A. Azouaoui**, C. Zouloumian, E. Garrabe, S. Doncieux. *Real-to-Sim: generating physically valid robot simulation scenes from text and images with LLMs and VLMs*. In preparation.

[2] **A. Azouaoui**, G. Nadizar, S. Cussat-Blanc, D. Wilson, M. Poette. *Interpretable prediction of clinician-rated respiratory severity in intensive care with genetic programming*. In preparation.